

Khai Yi Chin

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EXPERIENCE

Postdoctoral Scientist

June 2025 – July 2026

Amazon Robotics, North Reading, MA

- Designed a provably correct, decentralized self-assembly algorithm for a multi-robot system, validated across Monte-Carlo and Isaac Sim trials.
- Developed a distributed localization algorithm for a multi-robot system based on factor graph belief propagation.
- Developed RL locomotion policies that achieved stable rough terrain behavior, with sim-to-sim (Mujoco) and sim-to-real (DeepRobotics quadruped) validation.
- Established a containerized multi-GPU training workflow using Isaac Lab, Docker, and MLflow to scale parallel training experiments for quadrupedal locomotion development.
- Developed a self-reflecting, LLM-agent-orchestrated framework to improve autonomous robot recovery rates.

Graduate Research Assistant (Multi-Robot Sensor Degradation Estimation)

Sep 2021 – May 2025

NEST Lab at the Worcester Polytechnic Institute, Worcester, MA

- Created an extended Kalman filter-based recursive degradation estimation algorithm for flawed robot swarms.
- Built a Bayesian inference algorithm (Stein Variational Gradient Descent) in C++ for state estimation.
- Designed a minimalistic collective perception algorithm for robotic swarms with flawed sensors.
- Published work on collective state estimation using robot swarms (IROS, Swarm Intelligence (in review)).
- Extended the ARGoS multi-robot simulator with an EMANE network emulator to enhance simulation fidelity.
- Created simulation entities in C++ for robotic sensors and actuators in the ARGoS multi-robot simulator.
- Developed C++ drivers and controllers for the Khepera IV and AWS DeepRacer robots as testing platforms.

Robotics Engineer

Nov 2020 – Aug 2021

Coast Autonomous, Largo, FL

- Spearheaded development of a 3-D particle filter localization algorithm in C++, with an octree likelihood field.
- Achieved real-time localization along a >1,800 ft. outdoor trajectory in a >2 million sq. ft. map.
- Set up validation pipelines using ROS 2+Gazebo and RTMaps.
- Created deployment-ready, multi-platform (Windows and Linux) localization software libraries.
- Established the company's first robotics documentation infrastructure and version control system.

Autonomous Robotics Engineer (Autonomy and Simulation Lead)

Mar 2019 – Oct 2020

SIERA.AI, Austin, TX

- Led R&D project of an industrial autonomous mobile robot (AMR) with 10,000 lbs tugging capacity.
- Designed software and workflow for AMR deployment involving SLAM, localization, and autonomous navigation.
- Achieved ± 30 cm repeatability in infrastructure-free autonomous navigation of a >60,000 sq. ft. warehouse.
- Prevented AMR collisions during autonomous navigation through the integration of LiDARs and 3-D cameras.
- Improved AMR robustness by >30% through configuring, tuning, and implementing a fused odometry solution.
- Created software for performance benchmarking of different localization algorithms implemented on AMR.
- Enhanced AMR user experience by designing UI applications to provide robot and workflow management.
- Established complete simulation stack to streamline robotics software testing using AWS RoboMaker and ROS.

Mechanical Engineer (Product Development)

Apr 2018 – Feb 2019

DunAn Precision, Inc., Austin, TX

- Spearheaded mechanical design of 1st generation visual inertial measurement units (VIMUs) and test fixtures.
- Achieved in-house product assembly by implementing robotic and pneumatic systems.
- Improved VIMU accuracy by designing a sensor calibration program in MATLAB.
- Analyzed performance characteristics between ToF-based and IR stereo vision-based computer vision cameras.
- Investigated MEMS gyroscope designs using dynamical modeling in Simulink.

EDUCATION

Worcester Polytechnic Institute - *Ph.D. in Robotics Engineering*

2025

The University of Texas at Austin - *M.S. in Mechanical Engineering (Dynamic Systems & Controls)*

2018

University of Michigan, Ann Arbor - *B.S. in Engineering (Mechanical Engineering)*

2016

PUBLICATIONS

Multi-robot Systems

- **Chin, K. Y.**, Meng, T., Chen, Z., Bassett, D., and Ivanov, Y., 2026. "Huddle: Parallel Shape Assembly using Decentralized, Minimalistic Robots," *The 18th International Symposium on Distributed Autonomous Robotic Systems (DARS)*, Tokyo, Japan.
- **Chin, K. Y.** and Pinciroli, C., 2025. "BayesCPF: Enabling Collective Perception in Robot Swarms with Degrading Sensors," (In review); Preprint available on: [arXiv:2504.04774](https://arxiv.org/abs/2504.04774)
- **Chin, K. Y.** and Pinciroli, C., 2024. "Adaptive Self-Calibration for Minimalistic Collective Perception by Imperfect Robot Swarms," (In review); Preprint available on: [arXiv:2410.21546](https://arxiv.org/abs/2410.21546)
- **Chin, K. Y.**, Khaluf, Y., and Pinciroli, C., 2023. "Minimalistic Collective Perception with Imperfect Sensors," *2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, Detroit, MI.
- Akhter, A., Bhada, S. V., Golemi, K., Murphy, J., Wyglinski, A., Pinciroli, C. and **Chin, K. Y.**, 2022. "Towards Articulating Failures and Fault Trees and Verification for Building Resilience in Robot Swarms," *2022 IEEE International Symposium on Systems Engineering (ISSE)*, Vienna, Austria.

Soft Robotics

- Felt, W., **Chin, K. Y.**, and Remy, C. D., 2017. "Smart Braid Feedback for the Closed-Loop Control of Soft Robotic Systems," *Soft Robotics*, **4** (3), pp. 261-273.
- Felt, W., **Chin, K. Y.**, and Remy, C. D., 2016. "Contraction Sensing with Smart Braid McKibben Artificial Muscles," *IEEE/ASME Transactions on Mechatronics*, **21** (3), pp. 1201-1209.
- Felt, W., **Chin, K. Y.**, and Remy, C. D., 2016. "Self-Sensing Pneumatic Artificial Muscles for Feedback Control using the Inductance of 'Smart Braids,'" *Dynamic Walking 2016*, University of Michigan, Ann Arbor, MI.
- Felt, W., **Chin, K. Y.**, and Remy, C. D., 2015. "Dynamic Tracking of Joint Motion with Antagonized Smart Braids," *Fluid Power Innovation & Research Conference 2015 (FPIRC15)*, Chicago, IL.

Carbon Nanotube Modeling

- **Chin, K. Y.**, Cho, I., Zhang, J., and Fahrenthold, E. P., 2025. "Current Transmission in KAuBr₄ Doped Carbon Nanotube Wiring," *The Journal of Physical Chemistry C*, **129** (36), 16327-16337.
- **Chin, K. Y.** and Fahrenthold, E. P., 2021. "Mass specific performance of potassium tetrabromoaurate as a carbon nanotube dopant," *Computational Materials Science*, **197**, 110573.

SKILLS

Languages: C++, Python, MATLAB

OS: Linux, Windows

Software/Libraries: Eigen, PyTorch, Num/SciPy, Pandas, OpenMP, CppAD,

Isaac Sim/Lab, ROS (1 & 2), MLFlow, Gazebo, ARGoS, Git, Docker, SolidWorks

AWARDS

2x Amazon Inventor Puzzle Piece Award	March 2026
WPI Three Minute Thesis Competition - Honorable Mention	March 2025
WPI Robotics Engineering Graduate Student Travel Award	Jul 2023
WPI Graduate Research & Innovation Exchange (GRIE) - 2nd Place	Feb 2023
SIERA.AI Peer-to-Peer Reward and Recognition	Mar 2020
SIERA.AI Peer-to-Peer Reward and Recognition	May 2019
UT Austin Research Merit Fellowship	2018
Soft Robotics Toolkit 2015 Prize for Contributions in Soft Robotics Research - Winner	2015
University of Michigan Dean's List Award	Dec 2014
University of Michigan Dean's List Award	Apr 2014
University of Michigan Dean's List Award	Dec 2013